COMPUTER ENGINEERING

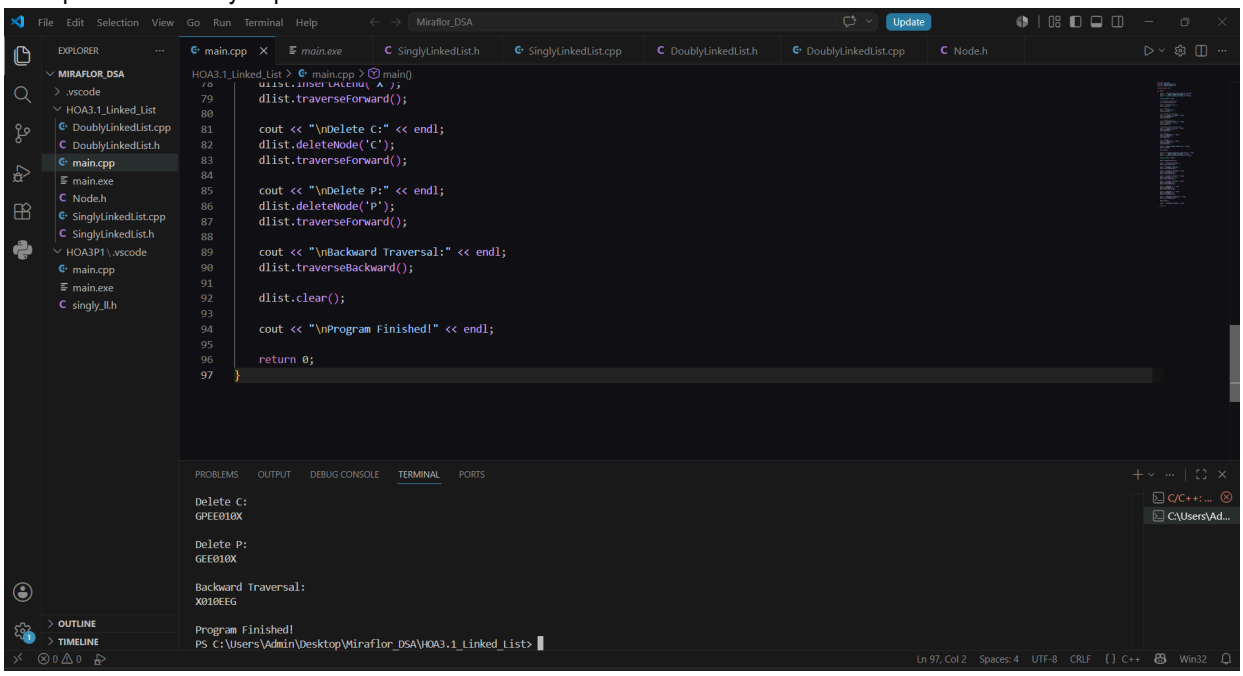
QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	26/07/2026	**DATE SUBMITTED:**	26/7/2026
NAME:	Seth Carlo T. Miraflor	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 3.1 Linked List

1. PROCEDURE OUTPUT

What procedures did you perform?



Discussion:

- The initial implementation creates a singly linked list representing the string CPE010. Each node stores one character and points to the next node. The last node points to NULL, indicating the end of the list. This implementation demonstrates how linked nodes are connected, but it is not efficient because every node is created manually. It can be improved by using functions to automatically create, insert, delete, and traverse nodes, making the code reusable and easier to maintain.

Traversal

```
Traversal: CPE010
```

Insertion at head

```
Insert G at Head:
GCPE010
```

Insertion at any part of the list

```
Insert E after P:
GCPEE010
```

Insertion at the end

```
Insert X at End:
GCPEE010X
```

Deletion of a node

```
Delete C:
GPEE010X

Delete P:
GEE010X
```

a. Source Code - list.traverse();
Console - CPE010
b. Source Code - list.insertAtHead('G');
list.traverse();
Console - GCPE010
c. Source Code - list.insertAfter('P', 'E');
list.traverse();
Console - GCPEE010
d. Source Code - list.insertAtEnd('X');
list.traverse();
Console - GCPEE010X
e. Source Code - list.deleteNode('C');
list.traverse();
Console - GPEE010X
f. Source Code - list.deleteNode('P');
list.traverse();
Console - GEE010X

Table 3.3

```
==============================
   DOUBLY LINKED LIST DEMO
==============================

Forward Traversal: CPE010
Backward Traversal: 010EPC
```

Analysis - The traversal operation visits every node from the head until the last node and displays each character stored in the linked list.

```
Insert G at Head:
GCPE010
```

Analysis - The insertion at the head creates a new node containing 'G' and makes it the new head of the list. The previous head becomes the second node.

```
Insert E after P:
GCPEE010
```

Analysis - The insertion after node 'P' adds a new node containing 'E' between P and the existing E, updating the links without affecting the remaining nodes.

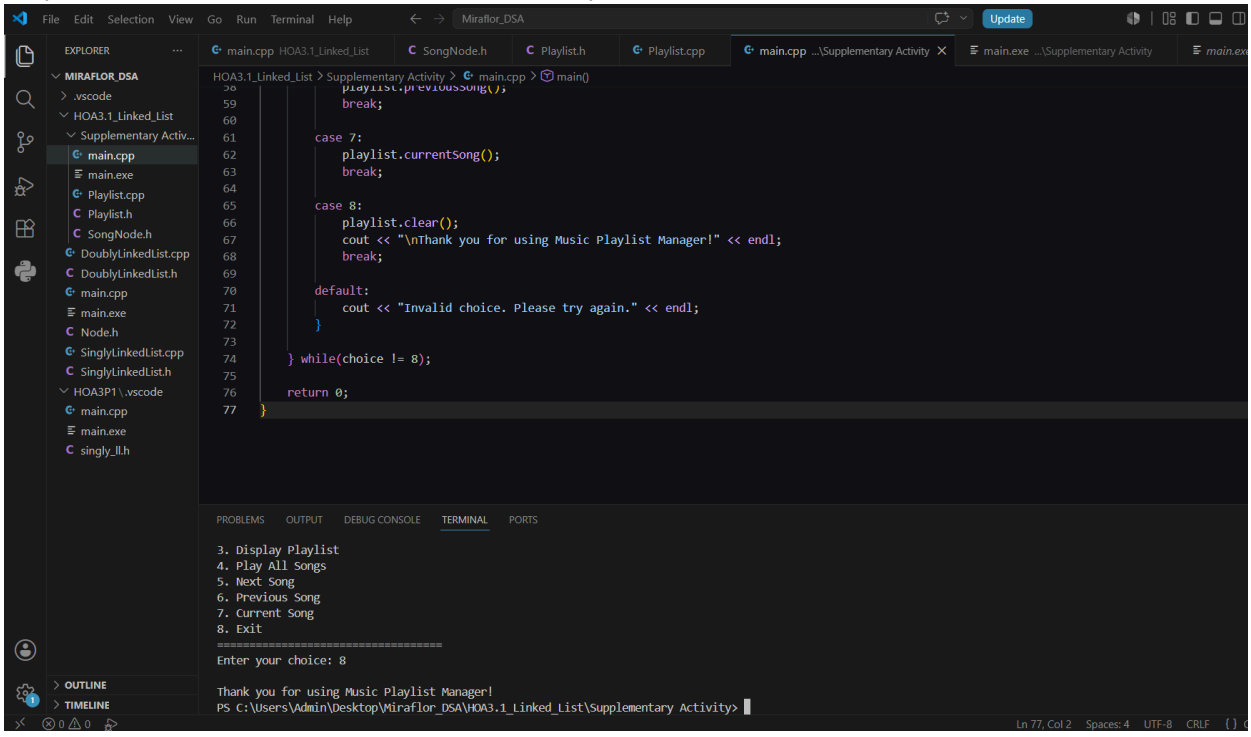
```
Delete C:
GPEE010X

Delete P:
GEE010X
```

Analysis - The deletion operation removes the specified node and reconnects the remaining nodes by updating the links. After deleting C and P, the final list becomes GEE010X.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.



3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work.

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The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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